Accelerated-Flow Wind-Tunnel Experimental Design: From Evidence Synthesis to a Synchronized Pressure and PIV Plan

Complete English technical review

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Abstract

Rectangular-cylinder response under a monotonic increase in freestream velocity depends on side ratio, attack angle, initial wake state, acceleration history, and the clocks of the measurement systems. The available literature does not support a single transient correction across these variables. This technical review links evidence classes and claim ceilings to a synchronized pressure and PIV design for three sections with $B/D = 1$, 3, and 5. The planned matrix contains 69 steady pressure combinations, 21 transient pressure angle magnitudes, 13 core PIV magnitudes, 23 signed PIV acquisition angles, and six locally smoothed target histories. UP is the primary planned branch. DOWN remains `DESIGN_ONLY_NOT_ACTIVE`. Native MATLAB generation and discrete-symmetry checks establish numerical consistency of the target files only. Formal analysis must replace nominal histories with measured Cobra $U(t)$, actual triggers, and recorded environmental properties. Pressure rate, PIV rate convention, controller tracking, synchronization, optical access, safety, and facility authorization remain explicit execution gates. The artifact is not_scientifically_approved and reports no project measurements.

This technical review is design material. It is not facility approval, controller-compatibility approval, safety approval, hardware authorization, public-release permission, or a report of measured wind-tunnel results. The governed literature corpus has `scientifically_admissible = 0 of 55`.

1 Introduction and literature review

1.1 The accelerated-flow problem and engineering relevance

Rectangular cylinders exposed to time-varying freestream velocity present a richer response space than steady-inflow aerodynamics alone can describe. Under a monotonic, smoothly ramped increase of freestream velocity, the surface-pressure field and near-wake of a two-dimensional rectangular cylinder may depart from the quasi-steady prediction. How that departure scales with the side ratio B/D, attack angle, initial wake state, and a non-dimensional acceleration parameter a_p remains incompletely resolved by the existing evidence base.

The practical consequences extend to wind-induced loads on structures during thunderstorm downbursts, gust fronts, and other transient atmospheric events where the inflow speed changes on a time scale comparable to the aerodynamic response time of the body. The engineering relevance motivates a careful experimental programme, but the scientific value depends on whether that programme can separate the competing influences of geometry, inflow history, measurement channel, and reference-state definition.

1.2 Steady rectangular-cylinder baseline and side-ratio dependence

The aerodynamics of rectangular cylinders under steady inflow has been investigated extensively. Okajima [Okajima(1983)] documented the variation of Strouhal number with width-to-height ratio across a wide range of sections. Knisely [Knisely(1990)] extended these measurements to include the effects of incidence angle, establishing the sensitivity of shedding frequency to geometric orientation. For square sections ($B/D = 1$), Vickery [Vickery(1966)] reported fluctuating lift and drag in smooth and turbulent streams, while Robertson et al. [Robertson et al.(1978)] and Huot et al. [Huot et al.(1986)] mapped surface-pressure distributions and identified the separation, reattachment, and vortex-formation regions on each face.

For elongated sections, the aerodynamics of the $B/D = 5$ rectangular cylinder has received sustained attention through the Benchmark on the Aerodynamics of a Rectangular 5:1 Cylinder (BARC). Bruno et al. [Bruno et al.(2014)] summarized the first four years of activity, highlighting persistent scatter in mean and fluctuating pressure on the side faces near reattachment. Mannini et al. [Mannini et al.(2017)] investigated the coupled effects of free-stream turbulence and angle of attack. Schewe [Schewe(2013)] examined Reynolds-number effects. Wu et al. [Wu et al.(2020)] combined experimental and numerical approaches at multiple angles. Guissart et al. [Guissart et al.(2022)] applied PIV to the $B/D = 5$ section at moderate Reynolds numbers. Da Matha Sant'Anna et al. [Da Matha Sant'Anna et al.(1988)] provided detailed pressure-field measurements for several rectangular cross-sections including intermediate side ratios. Physical evidence for $B/D = 3$ under any inflow condition remains comparatively sparse and depends primarily on a single physical benchmark (Yang and Mason [Yang and Mason(2019)]) supplemented by limited numerical studies.

1.3 Transient and accelerating-inflow evidence

The physical wind-tunnel experiments of Yang and Mason [Yang and Mason(2019)] constitute the principal monotonic-acceleration baseline. Using a 762 mm $\times$ 762 mm $\times$ 2.36 m open-jet suction tunnel, they measured pressure-integrated forces on rectangular cylinders under six acceleration histories (YM01 to YM06) with a_p spanning approximately 0.0009 to 0.0026. Their 30-repeat pressure ensemble documented delayed onset of wake instability followed by approximately five to six shedding cycles toward steady-flow aerodynamic behaviour. That 30-repeat count is a source-study benchmark and does not demonstrate convergence at 30 repeats for a different facility, different models, or revised histories.

Morello et al. [Morello et al.(2025a)] reported experiments on the aerodynamic behaviour of square and rectangular cylinders under accelerating inflow conditions. Morello et al. [Morello et al.(2025b)] examined time-varying freestream velocity effects on elongated rectangular cylinders. Zhou et al. [Zhou et al.(2025)] investigated $B/D = 5$ under transient accelerated and decelerated winds, providing evidence on both ramp directions.

The overshoot trend reported by Li et al. [Li et al.(2025)], in which aerodynamic coefficients of rectangular cylinders exceeded their steady-state values during acceleration, remains a testable numerical hypothesis. The physical benchmark (Yang and Mason [Yang and Mason(2019)]) reports no significant overshoot across its a_p span. This discrepancy between the numerical prediction and the available physical measurement defines one of the hypotheses that a synchronized experiment could test.

1.4 Time cells across facilities

Brusco et al. [Brusco et al.(2022)] identified constant-frequency time cells in the vortex shedding from a square cylinder in accelerating flows. These cells represent intervals of approximately constant shedding frequency separated by abrupt frequency jumps, a behaviour not predicted by quasi-steady models. Brusco et al. [Brusco et al.(2024)] extended this investigation to downburst-like accelerating flows. Lunghi et al. [Lunghi et al.(2024)] used LES to examine the influence of inflow acceleration on a square cylinder, providing numerical mechanism evidence for the formation and transition of time cells. These studies report time-cell behaviour under their respective facility and numerical conditions. Whether the phenomenon transfers across facilities, side ratios, angles, and acceleration histories requires systematic physical measurement.

1.5 Pressure-pattern deviation and reattachment sensitivity

The face-resolved transient pressure distribution, compared against a matched steady reference at the same instantaneous speed, reveals how the transient inflow reorganizes separation, reattachment, and vortex formation on each face of the section. For elongated sections, the reattachment position on the side faces is sensitive to angle, turbulence, and Reynolds number even under steady inflow (Bruno et al. [Bruno et al.(2014)]; Mannini et al. [Mannini et al.(2017)]; Patruno et al. [Patruno et al.(2016)]). Under accelerating inflow, the time required for reattachment to adjust to the instantaneous speed introduces a lag that differs among faces and among side ratios. The fine angle resolution required to capture reattachment migration on $B/D = 5$ is one of the design drivers for the dense 1°-step steady sweep at that section.

1.6 Numerical mechanism evidence and its status

Numerical studies provide mechanism hypotheses that physical experiments can test but cannot replace. Guo et al. [Guo et al.(2021)] investigated flow around a square cylinder in accelerated flow using numerical simulation. Lunghi et al. [Lunghi et al.(2024)] used LES for the square-cylinder case. Li et al. [Li et al.(2025)] predicted aerodynamic overshoot for rectangular cylinders. These numerical predictions are classified as simulation evidence (E-SIM) and enter the design as testable hypotheses rather than as established physical phenomena. The CFD, LES, DNS, and URANS results from Bruno et al. [Bruno et al.(2010), Bruno et al.(2012)], Chiarini and Quadrio [Chiarini and Quadrio(2021)], Mannini et al. [Mannini et al.(2011)], Ohya et al. [Ohya et al.(1992)], Oka and Ishihara [Oka and Ishihara(2009)], Sohankar [Sohankar(2008)], Ricci et al. [Ricci et al.(2017)], and others provide steady-inflow mechanism context but do not close the transient evidence gap.

1.7 The open questions this study addresses

The resulting gap is not simply a need for more conditions. It is a need for face-resolved pressure and flow-field measurements on a common coordinate system, a common measured freestream velocity history, a common trigger, and a common uncertainty basis, so that target-steady exceedance, instantaneous quasi-steady exceedance, and local-pressure overshoot remain distinct. Inflow families (monotonic ramp, accelerating and decelerating, downburst-like, streamwise sinusoidal, vertical sinusoidal, and steady turbulent) carry different time datums, directions, and normalizations and cannot be merged. This study derives a synchronized pressure and PIV experimental design that addresses these requirements for $B/D = 1$, 3, and 5 under monotonic ramp-up acceleration.

2 Scientific scope and accelerated-flow evidence base

2.1 Quantities of interest and the three exceedance definitions

The primary quantities of interest are surface pressure $C_p(t)$, pressure-integrated sectional coefficients $C_D(t)$, $C_L(t)$, and $C_M(t)$, wake-onset event times, time-cell boundaries and durations, PIV velocity and vorticity structures, the measured

freestream velocity $U(t)$, the achieved acceleration parameter, and the synchronization error between pressure and PIV channels.

Three non-interchangeable definitions of exceedance must be distinguished: (i) exceedance relative to the target steady-state value at the final speed U_t; (ii) exceedance relative to the instantaneous quasi-steady prediction at the current speed $U(t)$; and (iii) exceedance relative to a matched local steady measurement $C_{p,i}$ obtained at the same speed point during steady-baseline acquisition. Each definition answers a different question and may yield a different sign, magnitude, and duration. The experimental design must supply all three reference states.

2.2 Evidence classes and claim ceilings

Evidence entering the design is classified by provenance and constrained by a claim ceiling that prevents inadvertent upgrading. Table 1 summarizes the evidence classes. Physical wind-tunnel evidence (E-PHYS) supports a phenomenon to be reproduced. Digitized evidence (E-DIG) supports trend comparison under stated flags. Derived quantities (E-DERIV) require transparent formulae and inputs. Simulation evidence (E-SIM) supports a mechanism hypothesis. Compiled benchmarks (E-COMP) provide steady background references. Project design decisions (D-PROJ) report the current baseline. Facility specifications (F-SPEC) report datasheet capability and are not joint operating guarantees. Engineering QA results (Q-ENG) report numerical or structural checks.

The governed literature corpus contains 55 sources with `scientifically_admissible = 0 of 55`. Engineering closure of 52 sources and quarantine of 3 sources does not license controlling cross-source quantitative superposition.

2.3 Channel separation rules

Measurement channels do not substitute for one another. Local surface pressure is not a sectional force. A pressure-integrated force is not an independent balance force. Cobra and Pitot point velocity is not hot-wire turbulence measurement. A PIV velocity field does not supply total force. Each channel retains its native rate, uncertainty, and spatial coverage. No independent force balance is assumed in this design.

2.4 Inflow-family separation

Inflow families do not merge. Monotonic ramp, accelerating and decelerating, downburst-like (Li et al. [Li et al.(2022)]; S. Li et al. [S. Li et al.(2024)]; Brusco et al. [Brusco et al.(2024)]), streamwise sinusoidal (Li et al. [Li et al.(2021a)]; Wu et al. [Wu et al.(2021)]), vertical sinusoidal (Li et al. [Li et al.(2021b)]), and steady turbulent cases carry different time datums, directions, and normalizations. Only bounded parallel presentation or bounded trend comparison is permitted across families.

2.5 The $B/D = 3$ evidence position

$B/D = 3$ remains the weakest physical evidence position among the three target side ratios. The available physical evidence depends primarily on the single physical benchmark (Yang and Mason [Yang and Mason(2019)]). Numerical $B/D = 3$ mechanism results do not close the physical evidence gap. The design therefore treats $B/D = 3$ as a high-priority physical-evidence target.

Table 1: Evidence classes, permitted claims, and prohibited upgrades.

Class	Definition	Permitted claim	Prohibited upgrade
E-PHYS	Physical wind-tunnel result from a governed source	Report the phenomenon under the source conditions and channel	Not a result of this project; not transferable across inflow families
E-DIG	Value digitized from a published figure with QA flags	Redraw or compare trends with stated conditions and flags	Never presented as an author-reported table value
E-DERIV	Quantity computed from source tables or digitized points	Use with transparent formula, inputs, and comparison contract	Never presented as directly reported
E-SIM	CFD, LES, DNS, or URANS result	Mechanism, topology, or numerical trend, labelled as numerical hypothesis	Never fills a missing physical measurement
E-COMP	Compiled or database benchmark	Steady background reference	Never counted as an independent new experiment
D-PROJ	Project design decision, current baseline	Report as the current project design choice with its status	Never facility-approved, never executed, never measured
F-SPEC	Facility datasheet or capability statement	Report as facility-supplied capability at stated conditions	Never a model-position performance guarantee
Q-ENG	Engineering QA or verification result on project files	Report the numerical or structural check that passed	Never scientific approval, never facility approval

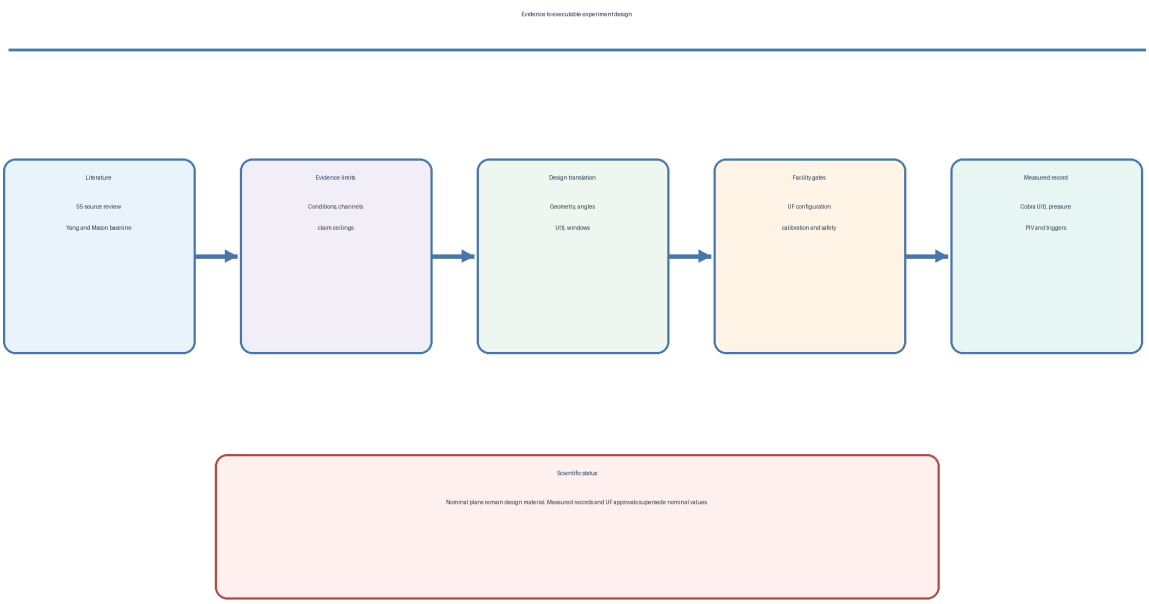


Figure 1: Evidence-to-experiment design chain: from evidence classes and constraint families to design decisions. Project-generated communication schematic.

3 From evidence to an executable experimental design

3.1 The eight constraint families

The shape of the experimental design is forced by the intersection of eight constraint families rather than chosen by preference.

First, the physical-modelling requirement and the scientific question fix three side ratios ($B/D = 1$, 3, and 5) at a common cross-flow depth, so that side-ratio comparison is not confounded by a changing reference length. Second, the literature supplies a protocol skeleton consisting of ramp-up plus hold, case labels, the a_p definition, and a 30-repeat pressure ensemble, but does not fix facility-specific parameters. Third, the facility-attainable speed range rewrites the source dimensional velocities into a UF-compatible endpoint set inside approximately 2 to 15 m/s while preserving each case label and its a_p. Fourth, instrument capability separates project requests from datasheet capability, which forces the pressure sampling rate and the PIV rate convention into open items. Fifth, the top-mounted camera and a vertically fixed field of view force signed attack angles, optical gates, and mechanism-selected windows instead of uniform PIV coverage. Sixth, synchronization requirements force native sampling rates, a common hardware trigger, and the honest label of locally synchronized or event-conditioned reconstruction instead of instantaneous global stitching. Seventh, storage and duty-cycle capacity produce the current PIV planning boundary and separate the complete physical clock, the PIV storage window, and the continuous pressure window into three distinct fields. Eighth, PI explicit decisions override earlier inferences and set the absolute clock anchors and the strict discrete symmetry requirement for the design-only descending branch.

3.2 Protocol selection: ramp-up plus hold

The protocol follows the Yang and Mason [Yang and Mason(2019)] baseline: a monotonic ramp from a low initial speed U_s to a target speed U_t, followed by a hold at U_t. The ramp is not an impulse, step, or sinusoidal oscillation. The six YM cases retain their original identifiers and case-specific a_p values while adopting UF-compatible speeds. This choice preserves comparability with the source benchmark while adapting to the host facility.

3.3 Endpoint rewriting under facility speed limits

The UF facility mean-speed range in the current configuration is approximately 2 to 15 m/s. The source Yang and Mason [Yang and Mason(2019)] endpoints (0 to 25.8 m/s in the fastest case) have been rewritten to UF-compatible pairs: U_s values of 2.3, 5.0, 8.3, and 11.0 m/s and U_t values of 11 and 15 m/s. Each rewritten case preserves the original a_p by adjusting the central acceleration a_{central} and the linear ramp time T_{linear}.

3.4 Decoupling the pressure matrix from the PIV matrix

The pressure and PIV matrices are decoupled. The 21 transient pressure angle magnitudes constitute a complete pressure-response survey. The 13 core PIV angle magnitudes and 23 signed PIV acquisition angles constitute a mechanism-targeted subset. PIV resource limits (storage, laser duty cycle, download time, and gantry repositioning) do not delete pressure evidence, and a PIV repeat count of 3 (up to 5 for critical cases) is not represented as a 30-repeat pressure ensemble.

3.5 The traceability rule from each decision to its driver

Each design decision must be traceable to its driver family. Physical-modelling constraints fix the models. Literature constraints set the protocol. Facility constraints set the endpoints. Instrument constraints identify open items. Optical constraints set the signed-angle convention. Synchronization constraints set the reconstruction label. Capacity constraints set the PIV planning boundary. PI decisions set the clock anchors and the DOWN-branch status. No design choice is attributed to a driver it does not have.

4 Planned models, coordinates, test matrix, and measurement channels

4.1 Three sectional models and the common reference depth

Three two-dimensional rectangular-cylinder models are planned. Model M11 has a 60 $\times$ 60 mm cross-section ($B/D = 1$). Model M31 has a 60 $\times$ 180 mm cross-section ($B/D = 3$). Model M51 has a 60 $\times$ 300 mm cross-section ($B/D = 5$). The common reference depth is $D = 0.060$ m (60 mm), defined as the cross-flow dimension at zero angle of attack for all three models. The nominal model span is 1200 mm. The typical wall thickness is 0.250 in. The model material is clear polycarbonate, supplied for installation; UF fabrication has not been requested. Pressure tubing length is 1.2 m.

4.2 Coordinate system, orientation, and attack-angle sign convention

At zero angle of attack, the long axes of M31 and M51 are streamwise. Left-to-right inflow defines the positive x direction. The pressure-face mapping uses Face I as the upstream vertical face, Face II as the upper face, Face III as the downstream vertical face, and Face IV as the lower face. The complete origin definition, positive-y direction, positive-angle convention, rotation centre, span, end condition, and support interface must be stated in controlled drawings issued by UF.

4.3 Pressure-tap and face mapping, channel budget

Draft tap totals are 26 for M11, 60 for M31, and 62 for M51, giving 148 total taps across all three models. The project pressure channel requirement is 64 channels with one model installed at a time. Corner taps are 0 on all three models.

Table 2: Model geometry, pressure channels, and draft tap distribution.

Property	**M11 ($B/D = 1$)**	**M31 ($B/D = 3$)**	**M51 ($B/D = 5$)**
Cross-section (mm)	60 $\times$ 60	60 $\times$ 180	60 $\times$ 300
Reference depth D (mm)	60	60	60
Span (mm)	1200	1200	1200
Wall thickness (in)	0.250	0.250	0.250
Tubing length (m)	1.2	1.2	1.2
Draft taps total	26	60	62
Face II taps	6 (1 to 6)	23 (1 to 23)	24 (1 to 24)
Face I taps	7 (7 to 13)	7 (24 to 30)	7 (25 to 31)
Face IV taps	6 (14 to 19)	23 (31 to 53)	24 (32 to 55)
Face III taps	7 (20 to 26)	7 (54 to 60)	7 (56 to 62)
Faces II/IV spacing	approx. 8.5714 mm	7.5 mm uniform	10 mm ends, five 20 mm centre
Faces I/III spacing	7.5 mm	7.5 mm	6, six 8, 6 mm chain
Corner taps	0	0	0
Spare channels (of 64)	38	4	2
Material	clear polycarbonate	clear polycarbonate	clear polycarbonate

The spare-channel arithmetic is tight: M51 leaves 2 spare channels and M31 leaves 4. The reference-pressure and diagnostic-channel budget (whether these sit inside or outside the 64 measurement channels) remains unresolved. These are draft scientific layouts, not manufacturing drawings or frozen channel maps. M31 physical millimetre coordinates are absent from the current CAD review, and no drawing carries a title block, revision, declared datum, or tap-to-channel map.

The following project CAD contact sheet places the face identifiers, inflow direction, coordinate axes, tap numbering, and 26/60/62 tap totals for all three sections in one controlled source figure. It is suitable as a draft scientific and model-layout figure. It is not a manufacturing release, frozen DAQ map, physical photograph, or measured result. The physical millimetre coordinates for M31 remain unresolved, and the title block, declared datum, and tap-to-channel map are not frozen. The complete contact sheet first preserves the source composition, followed by three content-preserving enlarged panels for article-scale readability.

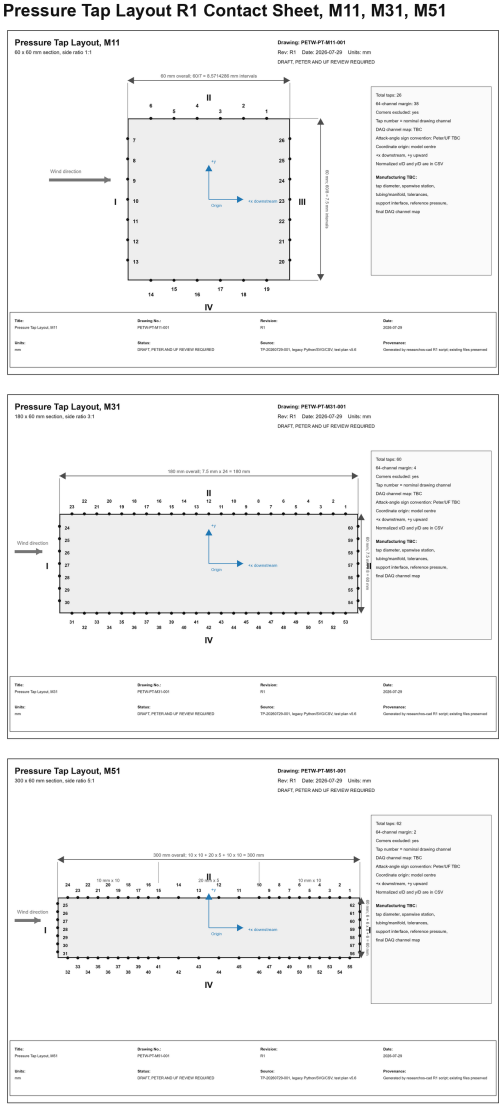


Figure 2: Project CAD pressure-tap layout for M11, M31, and M51, including sections, face identifiers, inflow direction, coordinate axes, tap numbering, and the 26/60/62 tap totals. It is suitable as a draft scientific and model-layout figure, not a manufacturing release, frozen DAQ map, physical photograph, or measured result.

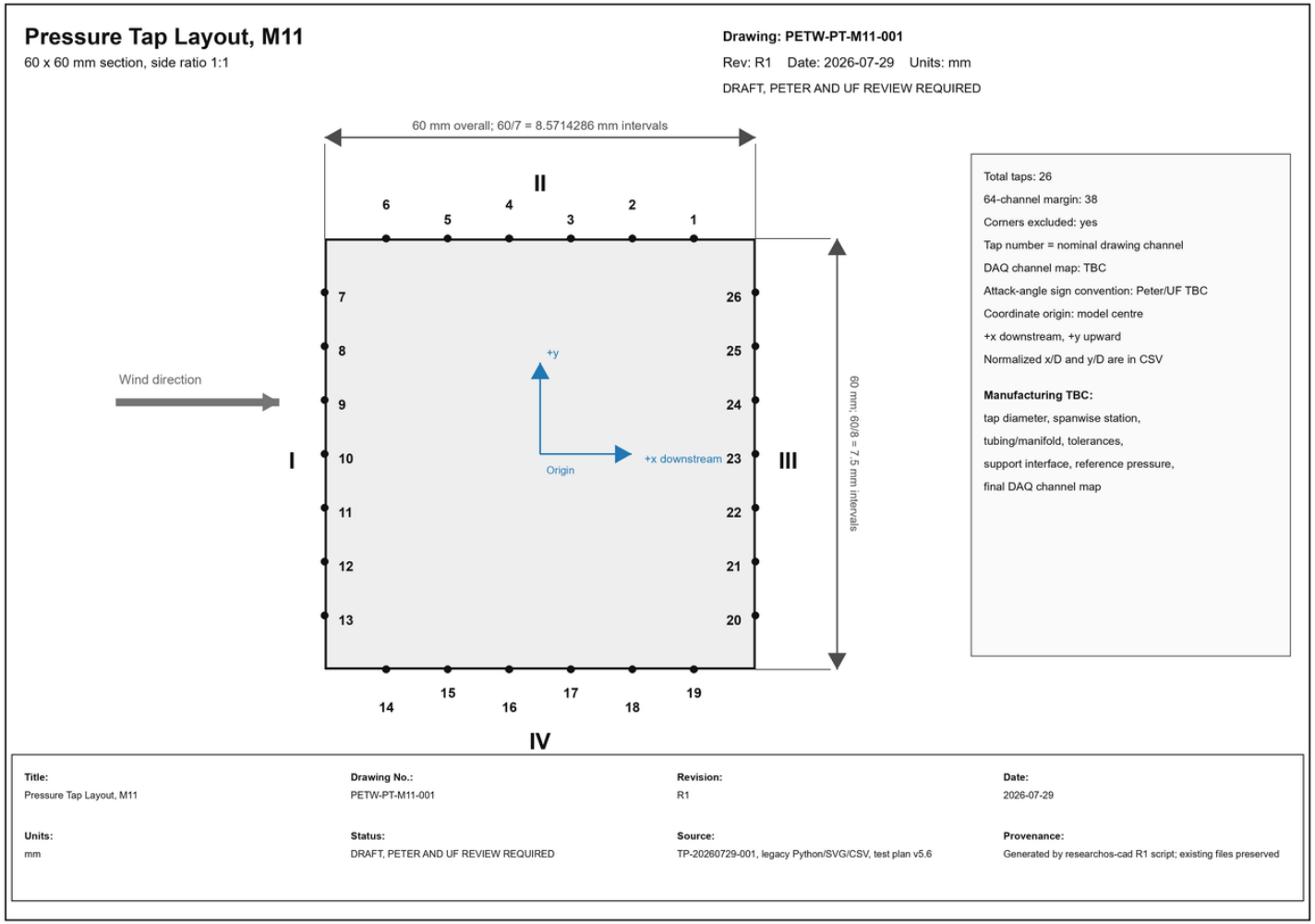


Figure 2 (continued), enlarged M11 panel. Status and interpretation limits are the same as Figure 2.

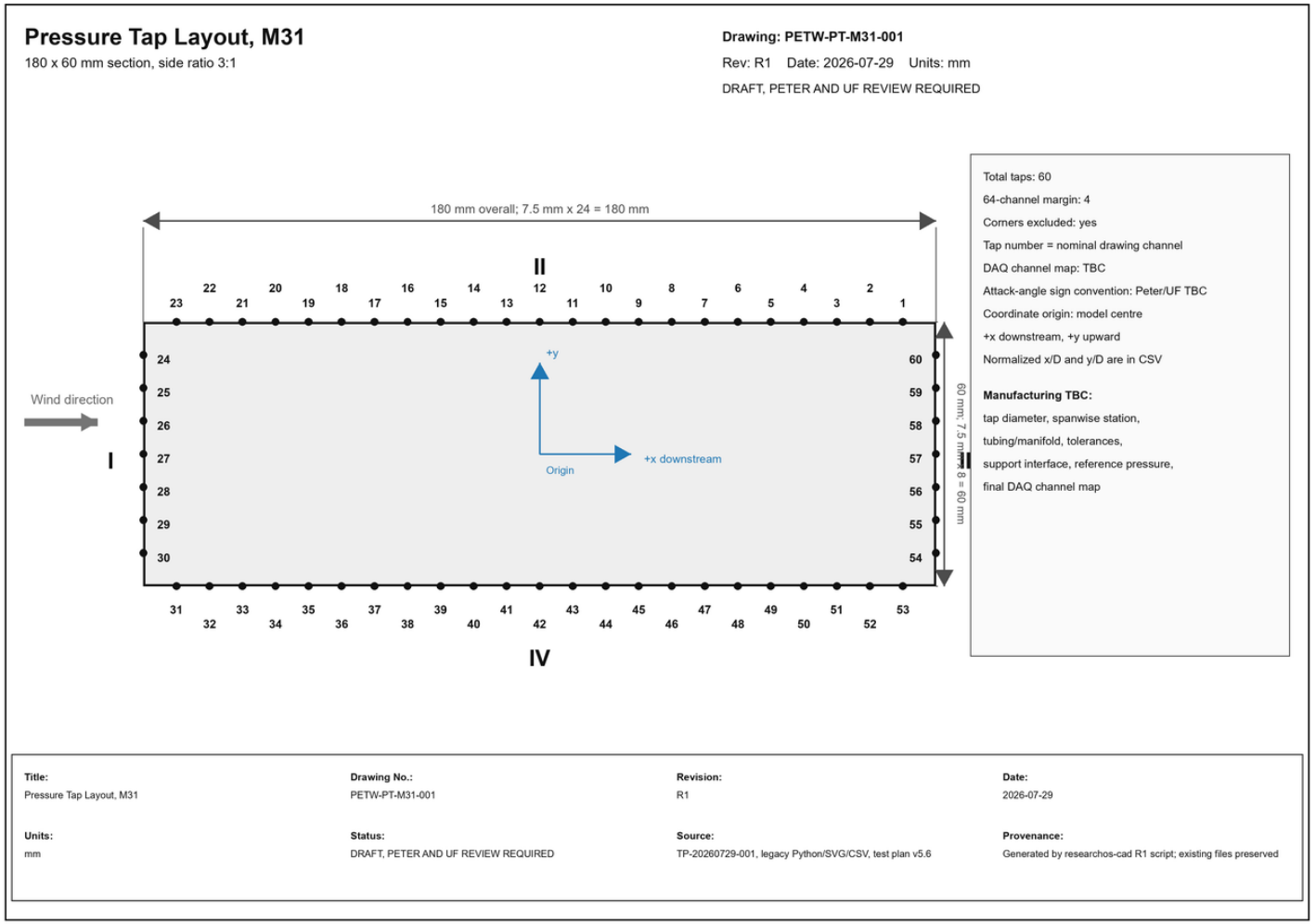


Figure 2 (continued), enlarged M31 panel. The physical millimetre coordinates, title block, declared datum, and tap-to-channel map for M31 remain unresolved.

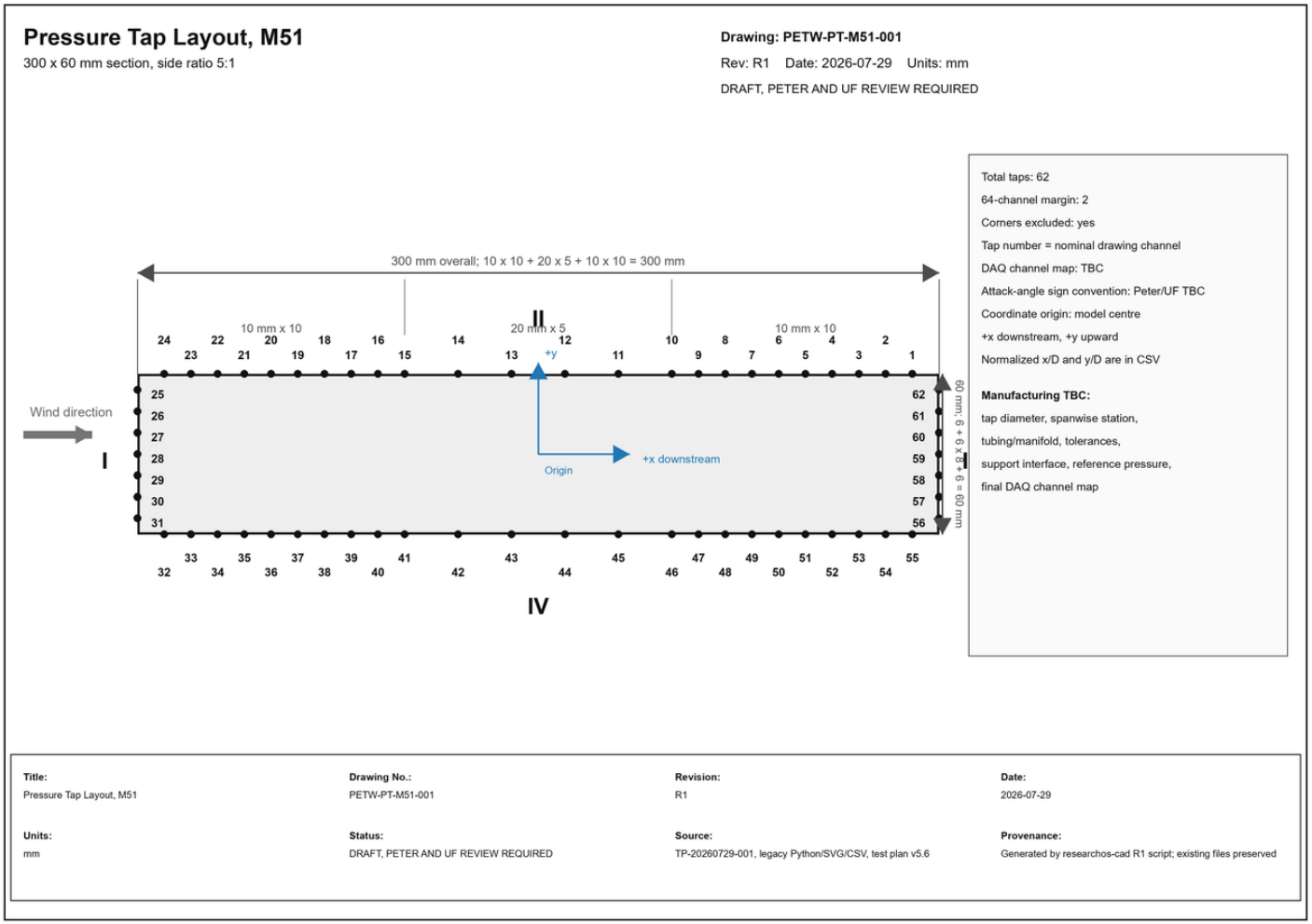


Figure 2 (continued), enlarged M51 panel. Status and interpretation limits are the same as Figure 2.

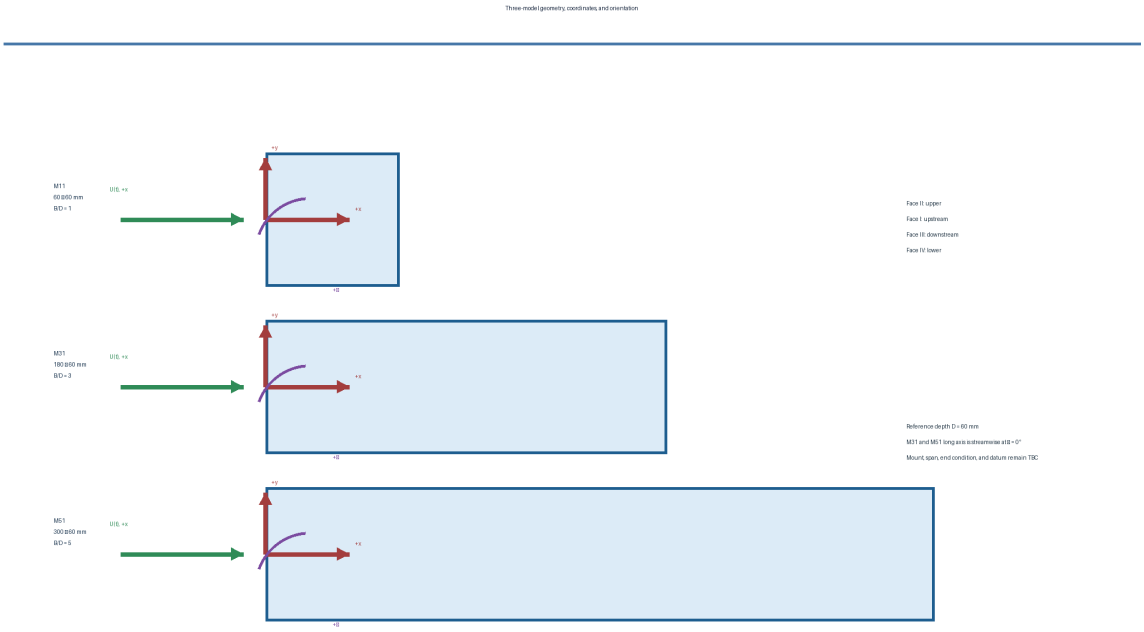


Figure 3: Three-model geometry and coordinate system: cross-sections, common depth, face naming, attack-angle sign convention, and draft tap distribution. Project-generated communication schematic.

4.4 Steady pressure matrix

The steady pressure matrix contains 69 model-angle combinations. Each combination requires at least 60 s of acquisition and at least 3 repeats, giving a minimum of $69 \times 3 = 207$ baseline collections before quasi-steady velocity points are added.

Table 3: Steady pressure matrix summary.

Model	Angle range	Step	Angle count	Min. acquisition	Min. repeats	Min. collections
M11	$-45°$ to $+45°$	$3°$	31	60 s	3	93
M31	$-18°$ to $+18°$	$3°$	13	60 s	3	39
M51	$-12°$ to $+12°$	$1°$	25	60 s	3	75
Total			**69**			**207**

Quasi-steady lookup construction requires 3 to 5 intermediate speed points spanning U_s to U_t per Tier 2 model-angle pair, establishing an independent speed-dependent reference that does not depend on the assumption that the steady value at U_t applies throughout the ramp.

4.5 Transient pressure matrix

The transient pressure matrix contains 21 angle magnitudes across the three models and six YM velocity histories.

Table 4: Transient pressure and PIV matrix summary.

Property	M11	M31	M51	Total
Transient pressure magnitudes	0, 3, 6, 9, 18, 27, 36, 45 (8)	0, 3, 6, 9, 12, 18 (6)	0, 1, 2, 3, 6, 9, 12 (7)	21
Core PIV magnitudes	0, 6, 18 (3)	0, 3, 6, 9, 12 (5)	0, 1, 2, 3, 6 (5)	13
Conditional PIV	27, 36, 45	18	9	–
Pressure-only by default	–	–	12	–
Signed PIV acquisition angles	5	9	9	23
Priority pairs	Pair A: YM01/YM03	Pair A: YM01/YM03	Pair A: YM01/YM03	–
	Pair B: YM02/YM04	Pair B: YM02/YM04	Pair B: YM02/YM04	–

The nominal Tier 2 planning baseline is $21 \times 6 \times 30 = 3{,}780$ runs. This is a planning baseline only; the final Tier 2 total is the sum of $N_p(\text{model}, \text{angle}, \text{case})$, which is not frozen. The signed-angle rule is that zero angle is acquired once, and each non-zero core magnitude is acquired at both $+\alpha$ and $-\alpha$.

4.6 PIV angle matrix and signed-angle logic

The PIV matrix contains 13 core angle magnitudes and 23 signed acquisition angles. The top-mounted camera views the model from above through a vertically fixed field of view ($y = 0$ to 200 mm, lower edge aligned with the model centreline at zero angle). Horizontal translation only is permitted. The three mechanism windows are W_{TE} (trailing edge and near wake, the default screening window), W_{LE} (leading-edge separation and reattachment), and $W_{\mathrm{W1+}}$ (downstream recovery, activated only when required).

The upper envelope of the model is

$$y_{\mathrm{top}} = 0.5\,[B\sin\alpha + D\cos\alpha]$$

The streamwise projection is

$$L_x = B\cos\alpha + D\sin\alpha$$

Signed-pair trailing-edge positions are

$$y_{\mathrm{TE,high}} = 0.5\,B\sin\alpha + 0.5\,D\cos\alpha, \qquad y_{\mathrm{TE,low}} = -0.5\,B\sin\alpha + 0.5\,D\cos\alpha$$

The signed-pair bottleneck margin is

$$m_{\mathrm{pair}} = \min\,[200 - y_{\mathrm{TE,high}},\; y_{\mathrm{TE,low}}]$$

Representative margins: M11 at 18° approximately 19.26 mm, at 36° 6.64 mm, at 45° 0 mm. M31 at 12° approximately 10.63 mm, at 18° 0.72 mm. M51 at 6° approximately 14.16 mm, at 9° 6.17 mm, at 12° -1.84 mm. The M31 streamwise projection at 18° is approximately 189.7 mm. The M51 streamwise projection from 0° to 12° ranges from approximately 300 to 306 mm, exceeding a single 200 mm window.

PIV repeats are 3 per dedicated window by default, with up to 5 for critical or unstable cases, subject to approval.

4.7 Mounting and end-condition assumptions

Model span (1200 mm), end plates, end condition, support, rotation centre, and blockage ratio at high attack angles remain subject to UF confirmation. Structural natural frequency relative to shedding frequency and rapid-ramp loading are safety items that must be resolved before G0 closure.

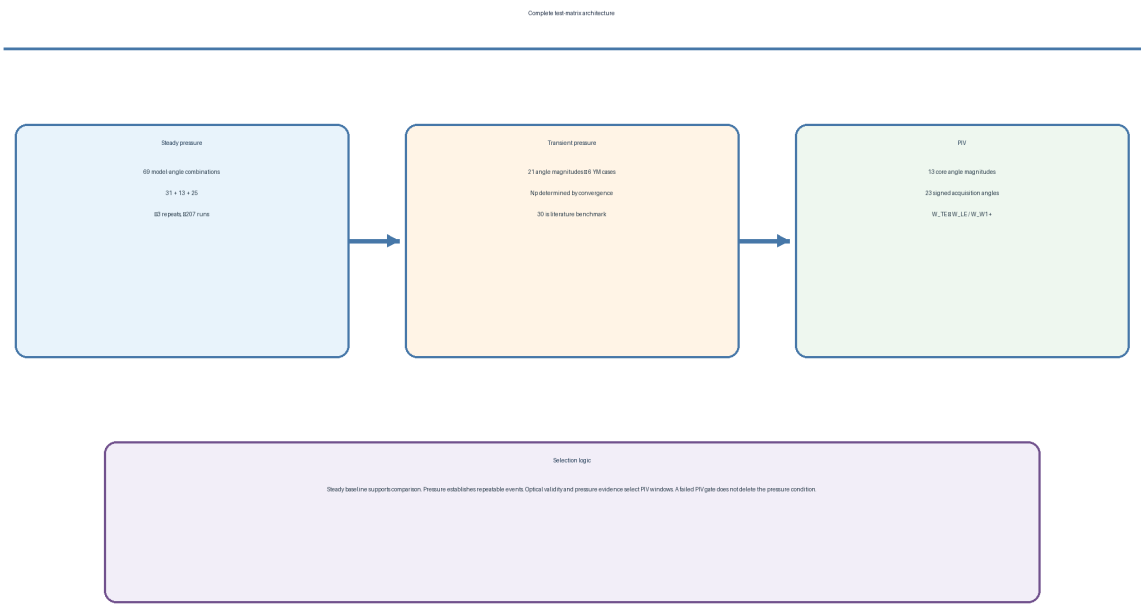


Figure 4: Complete test-matrix architecture with three lanes: steady pressure (Tier 1), transient pressure (Tier 2), and mechanism-oriented PIV (Tier 3). Counts 69, 21, 13, and 23 shown; conditional and pressure-only routes shown; totals marked provisional. Project-generated communication schematic.

5 Target velocity histories and UP/DOWN timing design

5.1 The six velocity cases and the acceleration parameter

Six velocity cases, YM01 through YM06, retain the Yang and Mason [Yang and Mason(2019)] case identifiers and case-specific a_p values while using UF-compatible start and target speeds within the approximately 2 to 15 m/s facility range. The non-dimensional acceleration parameter is defined as

$$a_p = \frac{a_{\text{central}} \cdot D}{U_t^2}$$

so that the central dimensional acceleration is

$$a_{\text{central}} = \frac{a_p \, U_t^2}{D}$$

The linear ramp time under constant central acceleration is

$$T_{\text{linear}} = \frac{U_t - U_s}{a_{\text{central}}}$$

The Reynolds number is defined as $Re = UD/\nu$ with nominal kinematic viscosity $\nu = 1.5 \times 10^{-5}$ m^2/s. These nominal Reynolds numbers are planning values only; formal analysis must use the measured freestream velocity $U(t)$, actual triggers, and recorded environmental properties to compute the achieved $Re(t)$ and a_p.

Table 5: Velocity cases, timing parameters, and nominal Reynolds numbers.

Case	U_s (m/s)	U_t (m/s)	a_p	a_{central} (m/s^2)	T_{linear} (s)	τ_s (s)	T_{rise} (s)	Re_s nominal	Re_t nominal
YM01	2.3	15.0	0.0012	4.500	2.822	0.564	3.951111	9,200	60,000
YM02	2.3	11.0	0.0026	5.243	1.659	0.332	2.322950	9,200	44,000
YM03	5.0	15.0	0.0012	4.500	2.222	0.444	3.111111	20,000	60,000
YM04	5.0	11.0	0.0025	5.042	1.190	0.238	1.666116	20,000	44,000
YM05	8.3	15.0	0.0010	3.750	1.787	0.357	2.501333	33,200	60,000
YM06	11.0	15.0	0.0009	3.375	1.185	0.237	1.659259	44,000	60,000

Case-pairing facts: YM01 and YM03 share $U_t = 15$ m/s and $a_p = 0.0012$. YM02 and YM04 share $U_t = 11$ m/s with close a_p values (0.0026 and 0.0025). YM05 and YM06 are the high initial-speed extension. U_s across the six cases spans 2.3, 5.0, 8.3, and 11.0 m/s.

The nominal shedding frequency at 15 m/s using a target Strouhal number $St = 0.13$ is approximately 32.5 Hz ($f_s = St \cdot U/D$). At a sampling rate of 300 Hz, this corresponds to approximately 9.2 samples per cycle.

5.2 Local Gaussian smoothing at the two slope transitions

Each target history applies local Gaussian convolution at the two slope transitions of an ideal top-hat acceleration profile. The smoothing half-width is fixed by

$$\tau_s / T_{\text{linear}} = 0.20$$

The Gaussian kernel width parameter is

$$\sigma_g / \tau_s = 0.35$$

The kernel is defined as

$$K(u) = \exp\left(-\frac{1}{2} \left(\frac{u}{\sigma_g \tau_s} \right)^2 \right), \quad u \in [-\tau_s, +\tau_s]$$

normalized to unit area over the truncation window. The effective rise time is

$$T_{\text{rise}} = T_{\text{linear}} + 2\tau_s = 1.40 \, T_{\text{linear}}$$

The velocity increment $\Delta U = U_t - U_s$ is preserved by the normalized kernel. The central quasi-linear portion retains approximately 72% of T_{linear} at the default parameters. Because the smoothed acceleration is a weighted average of 0 and a_{central}, the target command does not exceed a_{central} and introduces no artificial acceleration overshoot.

5.3 The UP physical clock and its three distinct windows

The UP branch uses a 20 s physical clock divided into three distinct windows. From 0 to 6 s, the model sits at the initial speed U_s with continuous pressure and Cobra acquisition but no PIV storage. From 6 to 8 s, PIV storage begins and captures a 2 s steady baseline at U_s. Acceleration begins at 8.000 s for all six cases. From 8 to 18 s (within the PIV storage window of

6 to 18 s), the velocity ramp and subsequent target-speed hold occur. From 18 to 20 s, the target-speed hold continues with pressure and Cobra acquisition, but PIV images are no longer stored.

The planning boundary $400\ \text{Hz} \times 12\ \text{s} = 4800$ frames is a capacity equality only. Whether 400 Hz denotes the camera frame rate, the image-pair rate, or the effective vector rate remains an unresolved open item that must be confirmed by UF.

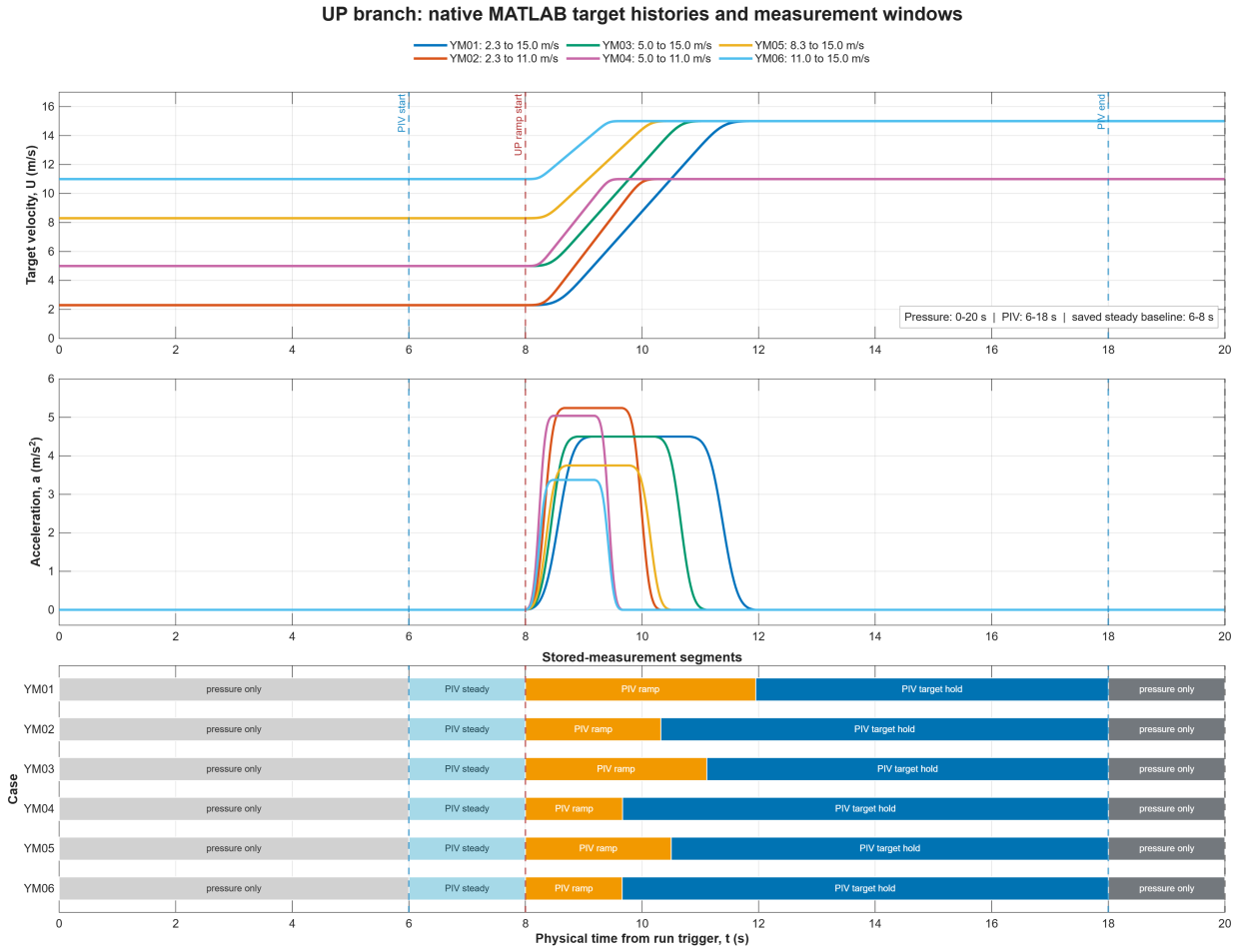


Figure 5: YM01 to YM06 UP target-velocity histories with acceleration and measurement segments. Native MATLAB authoritative plot produced by MATLAB R2026a through direct loading of the six native MAT files. These are design targets only; measured Cobra $U(t)$ supersedes target inputs in formal analysis. PIV window 6 to 18 s; common acceleration start 8 s; pressure window 0 to 20 s.

5.4 The design-only DOWN branch and strict discrete reversal (`DESIGN_ONLY_NOT_ACTIVE`)

The DOWN branch spans 0 to 33 s and remains `DESIGN_ONLY_NOT_ACTIVE`. It is a separate run with a separate trigger from UP. The initial steady segment at U_s extends from 0 to 8 s. The ascending ramp starts at 8.000 s (`DESIGN_ONLY_NOT_ACTIVE`). Each case holds at U_t until 22.000 s (`DESIGN_ONLY_NOT_ACTIVE`). Sampled hold before descent occupies 21.000 to 22.000 s, exactly 1.000 s inside the PIV window (`DESIGN_ONLY_NOT_ACTIVE`). Common deceleration starts at 22.000 s for all six cases (`DESIGN_ONLY_NOT_ACTIVE`). The deceleration, recovery to U_s, and remaining steady record span 22 to 33 s (`DESIGN_ONLY_NOT_ACTIVE`). The PIV storage window is 21.000 to 33.000 s (`DESIGN_ONLY_NOT_ACTIVE`), providing 1.000 s of PIV lead before the common deceleration point (`DESIGN_ONLY_NOT_ACTIVE`). The continuous pressure window is 0 to 33 s (`DESIGN_ONLY_NOT_ACTIVE`).

The descending segment is constructed as the strict discrete time reversal of the corresponding UP rise:

$$U_{\text{down}} = \text{flip}(U_{\text{up}}), \qquad a_{\text{down}} = -\text{flip}(a_{\text{up}})$$

This ensures identical velocity progression (reversed), sign-reversed acceleration, identical duration, and identical smoothing. The DOWN branch may only be considered after explicit PI and UF activation and closure of controller compatibility, run duration, storage, laser duty cycle, and safety gates (Gate G6).

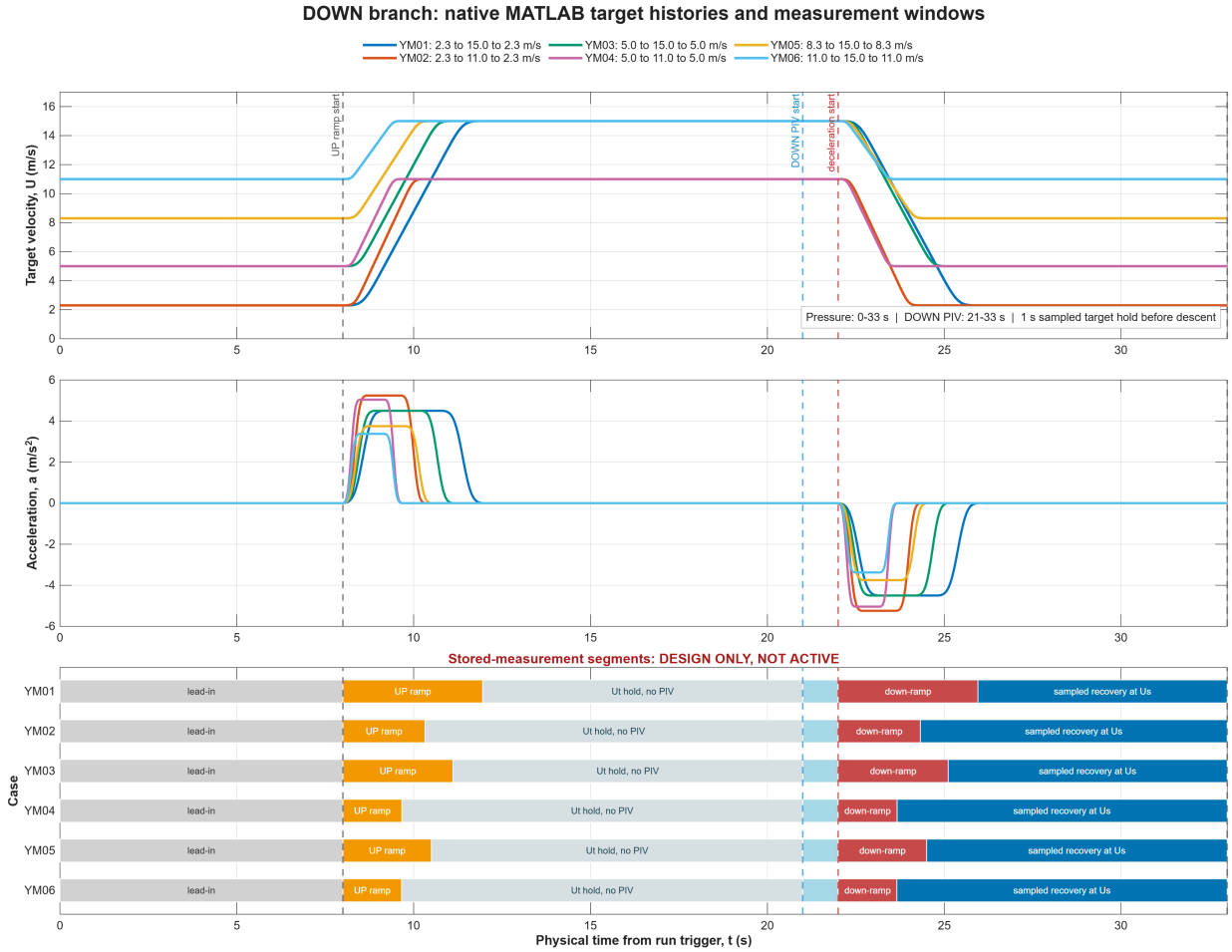


Figure 6: Strictly symmetric DOWN target-velocity histories with the 21 to 33 s PIV window and the 22 s descent anchor. `DESIGN_ONLY_NOT_ACTIVE`. Native MATLAB authoritative plot. No facility capability is implied. The 1 s lead before the common deceleration point provides a sampled hold reference within the PIV storage window.

Table 6: UP and DOWN clock anchors and window definitions.

Clock property	**UP**	**DOWN** (`DESIGN_ONLY_NOT_ACTIVE`)
Target-file coverage	0.000 to 20.000 s	0.000 to 33.000 s
Time step	0.001 s	0.001 s
File dimensions	20,001 × 2 per case	33,001 × 2 per case
Initial steady interval	0 to 6 s (pressure only)	0 to 8 s
Stored PIV baseline	6 to 8 s (2 s at U_s)	21 to 22 s (1 s at U_t)
Common ramp start	8.000 s	22.000 s (descent)
PIV storage window	6.000 to 18.000 s	21.000 to 33.000 s
Pressure window	0 to 20 s continuous	0 to 33 s continuous

5.5 Numerical symmetry verification

Independent QA for all six DOWN cases reports: maximum absolute velocity-symmetry error of 0 for all six cases, maximum absolute acceleration-symmetry error of 0 for all six cases, and T_{rise} and T_{fall} equal within floating-point representation (for example, YM01 3.952 s against 3.951999999999998 s). All six DOWN files contain 33,001 rows, all ending at 33 s, all with PIV window 21 to 33 s, all with 1 s lead. The overall QA status is PASS.

5.6 What the target files do not establish

The v6.7 target files and their numerical symmetry verification document native generation and internal consistency of the present target histories only. They do not establish UF controller compatibility, velocity tracking accuracy, DAQ synchronization, PIV rate convention, laser duty cycle, facility run-duration limits, or executability of the DOWN branch. The maximum MAT-to-CSV absolute difference of 5.0004×10^{-13} demonstrates cross-format numerical agreement but not controller interface agreement. The native environment is Windows MATLAB R2026a, version 26.1.0.3203278. The engineering check result string is `PASS_NATIVE_MATLAB_GENERATION_LOAD_AND_PLOT`. The MAT files use MATLAB v5/v7-compatible format, not v7.3 HDF5. The branch field values are `UP` and `DOWN_DESIGN_ONLY_NOT_ACTIVE`.

6 Pressure, PIV, synchronization, repeatability, and data quality

6.1 Pressure system and the sampling-rate open item

The facility pressure system is a 512-channel Scanivalve with a datasheet maximum of 625 Hz. The project pressure request is 64 channels at 1200 Hz with 1.2 m tubing. The 1200 Hz project request against the 625 Hz datasheet maximum remains an unresolved open item; 625 Hz must not be assumed to be the assigned per-channel rate for this experiment. Anti-aliasing configuration, reference pressure, trigger method, maximum record length, tubing transfer function, correction method, calibration procedures, zero and drift checks, leak checks, and the tap-to-channel map all remain UF-dependent items.

6.2 PIV optics, field of view, and mechanism windows

The facility PIV hardware is a Dantec stereoscopic PIV system with a DualPower 30-1000 laser (527 nm, 2×30 mJ at 1000 Hz) and two Phantom SpeedSense VEO 340 cameras (up to 72 GB, 4 MP, 800 fps per camera). The nominal PIV field of view is approximately 200 $\times$ 200 mm; the facility datasheet string `200 mm W x 200 m H` is treated as a suspected typographical error requiring written UF confirmation. The calibrated effective vector region of interest at the planned rate and stereoscopic configuration remains to be confirmed.

The vertical field-of-view placement is fixed at $y = 0$ to 200 mm, lower edge aligned with the model centreline at zero angle. Only horizontal translation is permitted. Three mechanism windows serve different scientific purposes: W_{TE} for the trailing edge and near wake (the default screening window), W_{LE} for leading-edge separation and reattachment, and $W_{\mathrm{W1+}}$ for downstream recovery, activated only when required. The M51 streamwise projection from 0° to 12° ranges from approximately 300 to 306 mm and exceeds a single 200 mm window, so one window cannot contain the full chord and wake simultaneously.

The following current project PIV planning figure combines the six velocity-case hierarchy, signed angle sets for the three models, mirror views at positive and negative attack angles, visibility margins, and the horizontal relationship among W_{TE}, W_{LE}, and $W_{\mathrm{W1+}}$. It preserves the fixed planning interval $y = 0$ to 200 mm and the horizontal-translation-only boundary. It is not a calibrated effective ROI, simultaneous multi-window measurement, facility approval, or measured flow field.

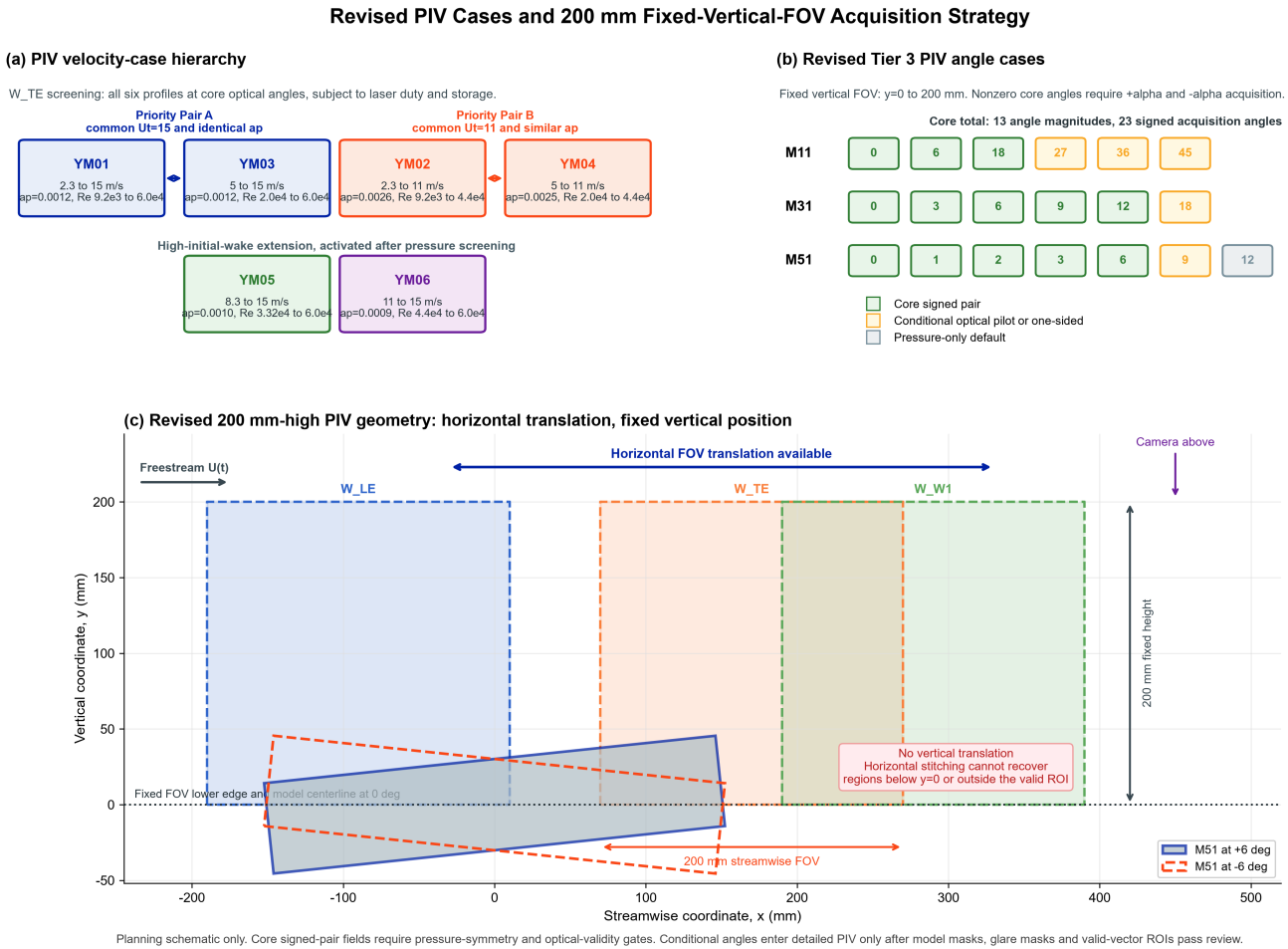


Figure 7: Current project PIV planning figure showing signed attack angles, positive/negative-angle mirror views, fixed vertical field of view, horizontal translation, visibility margins, and mechanism windows. The fixed planning interval is $y = 0$ to 200 mm, with horizontal translation only. It is not a calibrated effective ROI, simultaneous multi-window measurement, facility approval, or measured flow field.

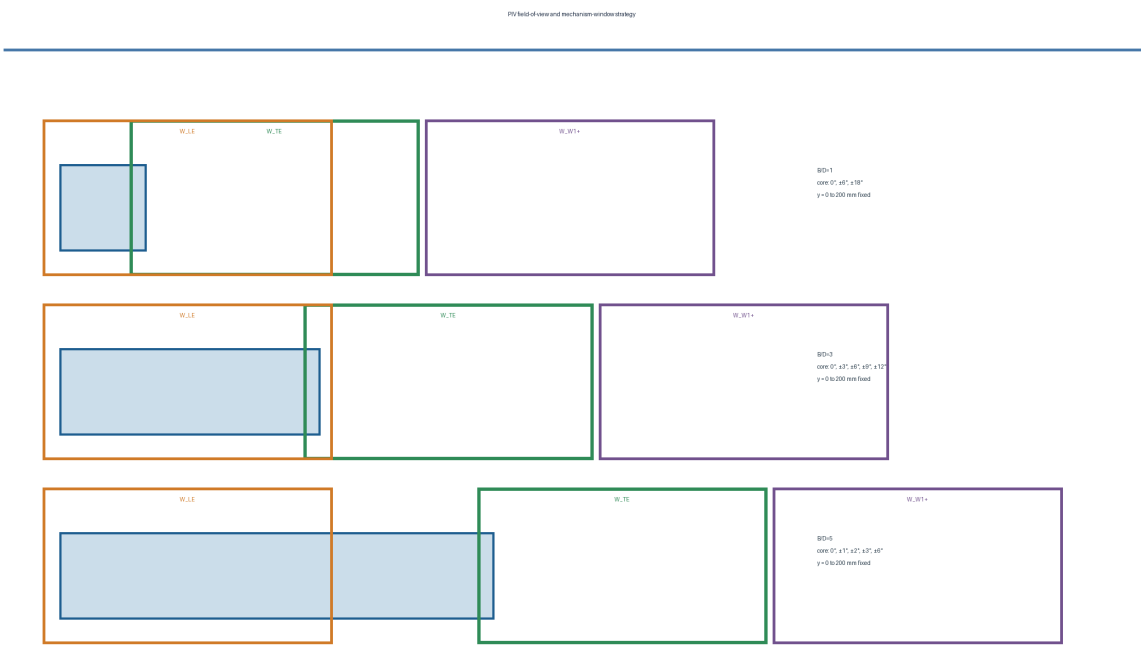


Figure 8: PIV field-of-view and mechanism-window strategy for $B/D = 1$, 3, and 5 at positive and negative attack angles. Fixed vertical extent $y = 0$ to 200 mm; horizontal translation only; nominal margins shown, not calibrated effective ROI. Project-generated communication schematic.

6.3 The capacity boundary and its three separate clocks

The PIV planning boundary is $400\ \text{Hz} \times 12\ \text{s} = 4800$ frames. The three distinct clocks are: (i) the complete physical clock (0 to 20 s UP; 0 to 33 s DOWN, `DESIGN_ONLY_NOT_ACTIVE`); (ii) the PIV storage window (6 to 18 s UP; 21 to 33 s DOWN, `DESIGN_ONLY_NOT_ACTIVE`); and (iii) the continuous pressure window (0 to 20 s UP; 0 to 33 s DOWN, `DESIGN_ONLY_NOT_ACTIVE`). These are separate fields and must not be conflated.

6.4 Freestream reference and the fixed Cobra requirement

One fixed tunnel-global TFI Cobra provides the common production freestream velocity $U(t)$. Gantry-mounted probes serve pretest surveys and QA only. The facility Cobra inventory consists of 3 TFI Cobra probes (up to 2000 Hz) and 12 synchronized Vectoflow Cobra probes (up to 850 Hz). No independent force balance is assumed; a balance remains an optional UF cross-check only. Pitot cross-check availability for mean speed remains to be confirmed. Measured Cobra $U(t)$ and actual triggers supersede all nominal target values in formal analysis.

6.5 Common trigger, native rates, and the reconstruction label

Each measurement system retains its native sampling rate. A common hardware trigger and shared master clock are the preferred synchronization approach. If the common trigger is not available, the clock mapping must be recorded and bounded. Joint analysis uses the effective PIV vector time as its grid; appropriately filtered pressure and Cobra data may be locally averaged or interpolated at those times, but upsampling cannot create information that was not sampled. The reconstruction is labelled as locally synchronized or event-conditioned reconstruction, never as instantaneous global field stitching.

The synchronized event-time relation distinguishes three time quantities: the trigger time (common to all channels), the native measurement time of each channel at its own sampling rate, and the PIV effective vector time (the time associated with a processed velocity vector from the double-exposure image pair). A bounded offset, skew, and drift between these times must be established at Gate G4.

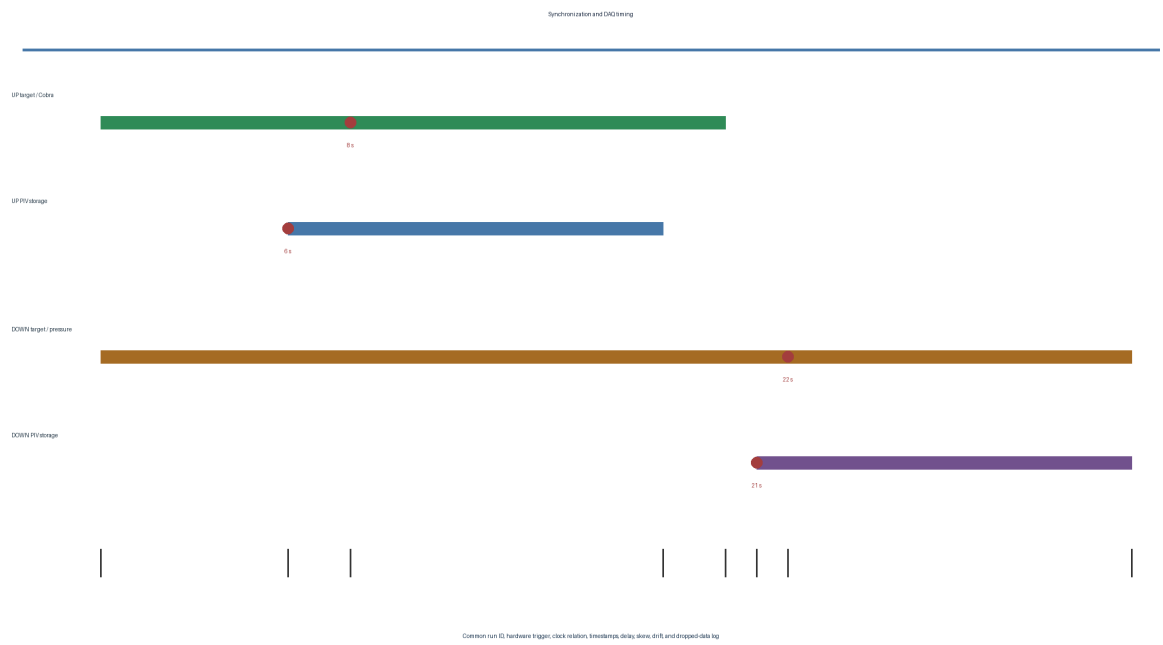


Figure 9: Synchronization and DAQ timing: common trigger, native rates, PIV vector effective time, and the three separate clocks. All rates shown are requested or TBC. Locally synchronized reconstruction stated. Project-generated communication schematic.

6.6 Pressure-repeat convergence against the 30-repeat benchmark

Pressure repeats are determined by a staged convergence pilot rather than fixed at 30. Cumulative ensembles are examined at 5, 10, 15, 20, 25, and 30 repeats for mean and RMS C_p, integrated $C_D/C_L/C_M$, exceedance magnitude and time, wake onset, time cells, measured $U(t)$, achieved a_p, and confidence intervals. PI and UF must confirm the stopping thresholds. The final N_p may equal 30, exceed 30, or be reduced after quantitative stability is demonstrated.

6.7 PIV repeat hierarchy

Dedicated PIV windows use 3 repeats by default and may increase to 5 for approved critical or unstable conditions. When events do not repeat at identical times, pressure-conditioned or phase-conditioned statistics are required. Unaligned images cannot be averaged and presented as an instantaneous mechanism.

6.8 Calibration, uncertainty, and acceptance checks

Table 7: Measurement subsystem requirements, completion conditions, and stop conditions.

Subsystem	Planning baseline / facility-dependent item	Completion condition	Stop condition
Pressure (Scanivalve)	64 ch, 1200 Hz request; assigned module, actual rate, anti-aliasing TBC	Traceable calibration, transfer function, zero/drift checks	Drift, saturation, leak, nonstationarity
PIV (Dantec stereo)	400 Hz × 12 s = 4800 frames; rate convention, camera config, laser duty TBC	Geometric calibration, valid-vector fraction, registration	Optical failure, drops, laser/storage limit
Cobra (freestream)	1 fixed global, gantry probes for QA; assigned system, mounting, angular range TBC	Native rate, bounded offset, continuous record	Missing trigger, unexplained clock error
Synchronization	Common trigger preferred; trigger method, clock mapping TBC	Bounded latency, skew, drift per channel pair	Missing trigger, unbounded skew or drift
Repeats (pressure)	5/10/15/20/25/30 convergence checks; PI stopping criterion TBC	Quantitative convergence demonstrated	Configuration change, nonstationarity
Repeats (PIV)	3 default, up to 5 approved; critical-case classification TBC	Conditioned statistics valid	Optical/storage/laser constraint
Uncertainty	All channels; tracking, calibration, repeat, PIV, timing	Bounded for every reported quantity	Unbounded component or missing raw data

7 Digital engineering architecture and reproducibility

7.1 Frozen-input and hash discipline

All frozen inputs are verified by SHA-256 hash at the start of each pipeline stage. The current source packet contains 18 files, all 18 verified with 0 mismatches and 0 missing files. Frozen inputs remain read-only throughout the pipeline. Target histories and their verification results are frozen design material, not measured data.

7.2 Native generation and independent read-back

The v6.7 target histories were generated natively in Windows MATLAB R2026a (version 26.1.0.3203278). Each UP file contains 20,001 rows ($\times$ 2 columns) at a 0.001 s time step. Each DOWN file (`DESIGN_ONLY_NOT_ACTIVE`) contains 33,001 rows ($\times$ 2 columns). The MAT-to-CSV maximum absolute difference is 5.0004×10^{-13}. The UP CSV values agree point-by-point with the prior verified numerical baseline, with zero time and velocity differences. Both figure PNG files are 3456×2652 pixels; both vector PDFs embed Arial and Arial Bold. The MAT variable set includes `target_velocity` ($N \times 2$), `time_s`, `speed_mps`, `acceleration_mps2`, `case_id`, `branch`, `Us_mps`, `Ut_mps`, `dt_s`, and `meta`.

7.3 Evidence-class separation carried into data products

The evidence-class separation defined in Section 2.2 is carried through into data products. Source-reported values retain their conditions and attribution. Project-confirmed planning values enter the current design. Facility-dependent values remain TBC until UF confirms them. Target $U(t)$ and measured $U(t)$ must be stored separately in all downstream products.

7.4 Run identity, metadata schema, and naming

The proposed run name encodes model, angle, condition, YM case, repeat number, measurement type, and window. Each run also records calibration IDs, channel map, gantry coordinates, nominal and measured U_s/U_t, measured segment boundaries, achieved a_p, environmental conditions, masks, valid-vector quality, exclusion reason, and raw-file links. Raw data remain read-only; processing scripts, parameters, and software versions are frozen with the results.

7.5 What the architecture does not authorize

The digital engineering architecture does not command, configure, or operate fans, the Flow Field Modulator, Terraformer, gantry, turntable, Scanivalve, Cobra, PIV cameras, lasers, interlocks, or DAQ. Qualified UF personnel retain all control and execution authority. No control file, gantry path, Terraformer input, or FFM command file is generated or authorized by this review.

8 Current readiness, unresolved questions, and staged execution gates

8.1 What is numerically complete

The UP target-velocity histories have native MATLAB generation and numerical verification (engineering check result `PASS_NATIVE_MATLAB_GENERATION_LOAD_AND_PLOT`). The strictly symmetric DOWN design (`DESIGN_ONLY_NOT_ACTIVE`) has independent QA verification (all six cases PASS). The UF technical prefill matrix contains 540 rows: 207 exact Tier 1 steady collections, 63 proposed steady 11 m/s endpoint lookup collections, 126 Tier 2 grouped model-angle-profile sequences, 138 conditional W_{TE} PIV candidate sequences, and 6 conditional W_{LE} and $W_{\mathrm{W1+}}$ placeholders. The workbook validation confirms exact nine-sheet order, zero external links, zero formula cells, one table per event sheet, and expected row counts passed.

8.2 What remains facility-dependent

All items in the pressure subsystem (assigned module, actual rate, anti-aliasing, reference pressure, trigger, record length, tubing transfer function, calibration, tap-to-channel map), PIV subsystem (rate convention, camera configuration, calibrated field of view, gantry coordinates, final case list), freestream reference (assigned Cobra, mounting, angular range), time base (common trigger, clock mapping, latency), flow quality (uniformity, turbulence intensity, spectra), controller (tracking of a_{central} and T_{rise} without saturation), duration (maximum run length, duty cycle, cooling), capacity (raw file sizes, download time, transfer, quota), and safety (blockage, structural frequency, emergency stop) remain facility-dependent.

8.3 The minimum pilot set

The minimum pilot requires one optically simple and scientifically relevant model-angle-case combination. The priority criterion is: (i) the model whose wake onset and time-cell behaviour are best predicted by the available literature, so that the pilot validates the entire measurement chain against a partially known expectation; and (ii) an angle where optical access is ample (large margin in the field-of-view calculation). The specific combination requires PI and UF agreement.

8.4 Stage gates from configuration confirmation to uncertainty analysis

Table 8: Staged execution gates.

Gate	Stage	Completion condition	Stop condition
G0	Configuration confirmation	PI and UF close channel, clock, controller-interface, model, mount, PIV, safety, and data-delivery fields	Any owner, value, authorization, or interface remains unresolved
G1	Flow-only check	Empty-section Cobra record covers 2 to 15 m/s and target tracking without unsafe behaviour	Tracking, flow quality, saturation, or safety criterion fails
G2	Steady baseline	Approved model and angle obtains calibrated pressure and Cobra baseline with repeatable statistics	Leak, drift, saturation, motion, unstable reference, or incomplete metadata
G3	Velocity reproduction	Measured Cobra $U(t)$ reproduces each accepted UP history within PI limits	Ramp boundary, overshoot, residual, cooling, or duration fails
G4	Synchronization pilot	Pressure, Cobra, PIV, and trigger timestamps yield bounded offset, skew, and drift	Missing trigger, unexplained latency, dropped data, or insufficient storage
G5	UP pilot	Selected model-angle-case completes optical, pressure-repeat, tracking, and data-integrity checks	Any production acceptance criterion fails
G6	DOWN decision (`DESIGN_ONLY_NOT_ACTIVE`)	PI and UF explicitly activate a reviewed, controller-compatible, safe branch	No explicit activation; status remains `DESIGN_ONLY_NOT_ACTIVE`
G7	Production acquisition	Frozen configuration completes approved matrix with live QA and immutable raw data	Configuration drift, failed QA, authorization loss, or storage anomaly
G8	Post-test QA	Run ledger reconciles files, channels, calibration, exclusions, and repeat convergence	Missing raw data, untraceable exclusion, bad calibration, or unreconciled run
G9	Uncertainty analysis	Reported quantities include synchronization, calibration, repeat, PIV, and tracking uncertainty	Uncertainty cannot be bounded, or measured $U(t)$ cannot replace nominal values

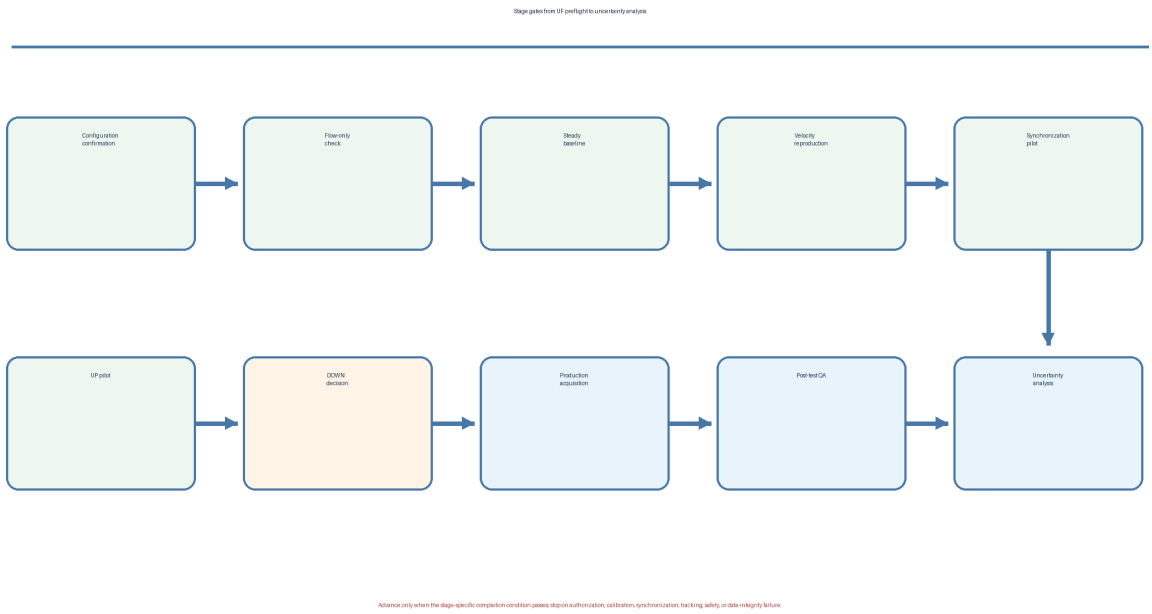


Figure 10: Stage-gate diagram from G0 (configuration confirmation) to G9 (uncertainty analysis). `DESIGN_ONLY_NOT_ACTIVE` DOWN decision gate shown as a separate, currently inactive branch. Project-generated communication schematic.

Table 9: Current open decisions and unblocking conditions.

Open item	Owner	Current status	Unblocking condition
Pressure rate (1200 Hz request vs. 625 Hz max)	UF	Unresolved	UF confirms assigned module and rate
PIV 400 Hz definition	UF	TBC	UF confirms frame rate vs. pair rate vs. vector rate
M31 tap coordinates	PI	Missing from CAD CSV	Dimension-controlled drawing issued
M51 spare channels (2 of 64)	PI	Tight	Reference/diagnostic channel budget resolved
DOWN activation	PI	`DESIGN_ONLY_NOT_ACTIVE`	Gate G6 closure
Controller tracking	UF	Not demonstrated	Gate G3 closure
Final N_p	PI	Not frozen	Convergence pilot and stopping rule
Conditional angle downgrading	PI	Not decided	Priority order for resource reduction
UF admin fields (UPN, dates, PI)	UF	Open	Administrative confirmation
Workbook visual review	PI	LibreOffice could not load	Microsoft Excel review completed
Yang and Mason point count (15,517 vs. 15,518)	PI	Unverified	Reconciliation manifest authorized
Bibliography extension for mechanism sources	PI	Not decided	PI authorizes additions
Scientific admissibility adjudication	PI	0 of 55	Adjudication scope defined

8.5 Completion and stopping conditions at each stage

Advancement to the next gate occurs only after the stage completion condition passes. Missing authorization, tracking failure, calibration failure, synchronization drift, sensor saturation, model motion, laser or facility limits, and raw-data anomalies trigger a stop. No gate may be bypassed. The DOWN decision gate (G6) remains a separate, currently inactive branch; the status `DESIGN_ONLY_NOT_ACTIVE` persists until PI and UF jointly and explicitly activate a reviewed, controller-compatible, and safe descending branch.

The Flow Field Modulator datasheet figures of 23 m/s free-discharge speed and 100 m/s^2 peak acceleration are facility-supplied capability at stated conditions and are not model-position performance guarantees. The facility data infrastructure includes a 10 Gbps network and DesignSafe archiving. The minimum number of model events is 3 (one per model); the final count remains TBC. The requested turntable envelope is $-45°$ to $+45°$ overall, with narrower model-specific ranges.

9 Discussion

9.1 What the design can and cannot discriminate

The scientific value of the design is that side ratio, initial wake state, and attack angle enter a common measurement protocol while evidence channels retain their identities. The 69 steady combinations provide baselines for transient exceedance and quasi-steady interpolation. The 21 transient-pressure magnitudes are planned to assess event repeatability. The 13 core PIV magnitudes and 23 signed angles explain mechanisms where optical access permits. Decoupling pressure and PIV prevents PIV resource limits from deleting pressure evidence and prevents a small PIV repeat set from being represented as a 30-run pressure ensemble.

The three exceedance definitions (target-steady, instantaneous quasi-steady, and matched local steady) can each be evaluated independently because the steady-baseline matrix provides the required reference states at multiple speed points. The time-cell analysis can proceed on the pressure records alone, with PIV providing mechanism context at selected times and angles rather than continuous spatial coverage.

9.2 Sensitivity of conclusions to the open facility items

The actual pressure sampling rate may be below the 1200 Hz project request. If the assigned rate is 625 Hz or lower, the maximum resolvable frequency, the number of samples per shedding cycle, and the time resolution of exceedance events all change. The PIV rate convention affects the temporal resolution of mechanism snapshots and the total number of usable vector fields per run. The controller tracking accuracy determines whether the achieved a_p and T_{rise} match the target values closely enough to preserve comparability with the Yang and Mason [Yang and Mason(2019)] conditions. These items are facility-dependent and cannot be resolved by design alone.

9.3 Relation to the physical benchmark and to the numerical hypothesis

The Yang and Mason [Yang and Mason(2019)] benchmark establishes the delayed-onset and time-cell phenomena for square cylinders under monotonic acceleration. The present design extends the investigation to $B/D = 3$ and $B/D = 5$ under a common protocol with face-resolved pressure and mechanism-targeted PIV. The overshoot hypothesis (Li et al. [Li et al.(2025)]) can be tested directly once measurements exist. If the physical measurements show no overshoot at the present a_p values, the hypothesis is bounded from above; if overshoot appears, its dependence on side ratio and angle can be characterized. Neither outcome can be anticipated by the design.

9.4 Limitations

Clear limitations remain. M31 tap coordinates are not dimension-controlled. M51 leaves only 2 of 64 pressure channels spare. Model span and end conditions are not frozen. The DOWN branch is numerically symmetric but has not been tested for controller compatibility, safety, or facility feasibility. Reproducible target generation does not demonstrate reproducible flow. Only measured $U(t)$, trigger records, pressure, PIV, and calibration records can support post-test scientific judgment. The governed literature corpus has `scientifically_admissible = 0 of 55`, and no cross-source quantitative superposition is licensed by the current evidence governance.

10 Concluding remarks

The available evidence supports a gated, synchronized pressure and PIV experiment on rectangular cylinders under monotonic accelerating inflow, but it does not support one transient correction across $B/D = 1$, 3, and 5. The current design places three models (60 × 60, 60 × 180, and 60 × 300 mm), 69 steady combinations, 21 transient-pressure magnitudes, 13 core PIV magnitudes, 23 signed acquisition angles, and six YM target histories on a common coordinate, velocity, and trigger basis. Local Gaussian smoothing with $\tau_s/T_{\text{linear}} = 0.20$ and $\sigma_g/\tau_s = 0.35$, the UP clock from 0 to 20 s with a common acceleration start at 8 s, the UP PIV window from 6 to 18 s, and the strict DOWN reversal (`DESIGN_ONLY_NOT_ACTIVE`) with a PIV window from 21 to 33 s are all traceable design materials whose formal analysis must use the measured Cobra freestream $U(t)$, actual triggers, and recorded environmental properties.

Completion requires all ten gates, G0 through G9: configuration confirmation, flow-only check, steady baseline, velocity reproduction, synchronization pilot, UP pilot, DOWN decision, production acquisition, post-test QA, and uncertainty analysis. No gate may be bypassed. This review remains not_scientifically_approved and does not constitute facility approval, controller-compatibility approval, safety approval, hardware authorization, public-release permission, or measured wind-tunnel results.

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Boundary. This draft is not_scientifically_approved. It is not UF facility approval, controller compatibility approval, safety approval, hardware authorization, public-release permission, or a report of measured wind-tunnel results.